10) Wet rivet fins where applicable,

11) Assemble in fixture with bonding materials and bond,

12) Automatic rivet flat areas between end closures as required,

13) NC machine edge trim, attachment holes, fin and secondary bead edge margins,

14) Clean.

23.1.1 Circular Tube Panel Assembly

Forming of the 0.025 inch thick beaded circular tube panels is described in Section 22. Figure 23-1 shows one panel, an edge doubler and a fingered end doubler being prepared for bonding. Figure 23-2 shows the two panels, the edge doubler, the fingered end doubler and an external end doubler being prepared for bonding. The internal fingered doublers were scalloped slightly smaller than the inside mold line of the bead end closure. The end three inches of the doubler fingers were chem mill tapered from 0.025-inch to approximately 0.005-inch. The fingered doublers were required to provide shear continuity in the ends of the panels and to transmit the loads into the ends of the beads. The long narrow external doublers were required to provide increased bearing area for the mechanical fastener attachments at the edges of the panels.

The bonding materials consisted of the phenolic primer type I #EC-2320, a modified epoxy type II #AF-126, class I, grade 5 and a positioning fabric, produced by the 3M Company. The bonding was conducted per BAC Specification 5514. The autoclave cycle for these materials is one hour at 250°F with a bond line pressure of 25 to 100 psi.

Figure 23-3 shows a bonded panel with the bonding fixture. The straight elements of the bonding fixture were rectangular aluminum bars and the curved end members were cut from thick aluminum plate using a band saw.